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Galway and Roscommon Education and Training Board

Maths for Trades

MATHS FOR METAL FABRICATION APPRENTICES

Phase 1 & Phase 2

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Maths for Trades

MATHS FOR METAL FABRICATION APPRENTICES

By Thelma Donoghue

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Contents

About this Book

The ETBI Support to Apprentices Group is a subgroup of the Directors of Further Education and Training Forum comprising of staff who support apprentices with literacy, numeracy, study skills and IT, and includes representatives from all 16 ETBs.

The group is primarily a community of practice, sharing resources, online materials and best practice focusing on Phases 1 & 2 Craft Apprenticeships

The first series comprises:

- Maths for Carpentry & Joinery Apprentices
- Maths for Commis Chef Apprentices
- Maths for Electrical Apprentices
- Maths for Metal Fabrication Apprentices
- Maths for Motor Mechanics
- Maths for Plumbing Apprentices

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Siobhan McEntee & Alison Jones, Co-Chairs, ETBI Support to Apprentices Group

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Foreword

ETBI and GRETB are delighted to publish the first series of 'Maths for Trades' handbooks for Phase 2 Apprentices in Further Education and Training. Education and Training Boards offer maths support to apprentices during Phase 1 and Phase 2 to equip apprentices with the mathematical knowledge required for their individual apprenticeship.

The Support to Apprentice Programme, a nationwide initiative, is progressive in terms of educational methodologies, assessing maths skills and providing tailored programmes to support craft apprentices to achieve their full potential.

Education and Training opportunities are constantly evolving to meet the skill gaps of employers and consequently FET must adapt and enhance programmes and modes of delivery to meet this demand. Creating content and resources such as Maths for Trades ensures that ETB FET learners get the best opportunity available to them in their chosen area of learning.

The Support to Apprentice Group have produced the first range of Maths for Trades handbooks based on identified needs of apprentices in collaboration with instructors and tutors in ETB Training Centres and Literacy Services.

This is an exciting time in education and a great opportunity for Further Education and Training managers, teachers, tutors, instructors and coordinators to participate in the creation and roll-out of focused, contextually based resources.

Eithne Nic Dhonnchadha

Director of Further Education and Training

Galway and Roscommon ETB

Nessa White

General Secretary

Education and Training Boards Ireland

Unit 1: Arithmetic & Calculator Use

We can use a calculator for all sections of this course. Using a calculator effectively is not simply a matter of pressing the right key; we must learn some key mathematical rules as well.

1.1 Addition & Subtraction

Complete the following exercise:

A. 11208 + 220.7 =	**B.** 79 + 39 + 293 =
C. 685.8 + 108.5 =	**D.** 212 + 963 + 465 =
E. 378.67 + 212.44 =	**F.** 69 – 21 =
G. 65 + 28 + 112 =	**H.** 5706 + 2012 =
I. 236 – 116 =	**J.** 38.6 - 16.8 =

1.2 Multiplication & Division

Complete the following exercise:

A. 458 x 326 =	**B.** 688.25 ÷ 38.91 =
C. 268.6 x 29 =	**D.** 2120.25 ÷ 16 =
E. 389.21 x 38.6 =	**F.** 3165.98 ÷ 57 =
G. -15 x 26 =	**H.** -112.5 ÷ 25.4 =
I. 65 ÷ 5 =	**J.** 177.7 ÷ (-16.5) =
K. 128 ÷ 4 =	**L.** 127 x 25 =

1.3 Squares, Square Roots & Indices

We can use a scientific calculator to find squares and square roots but first what are they?

When we multiply a number by itself the result is called the square of the number.

Square of a Number:

Example:

- 2 squared = 2 x 2 = 2^2 = 4
- 3 squared = 3 x 3 = 3^2 = 9
- 4 squared = 4 x 4 = 4^2 = 16

When using the calculator, we need the square key $\boxed{x^2}$ to find the square of a number.

Example:

- 26^2 = 26 $\boxed{x^2}$ = 676
- 54.3^2 = 54.3 $\boxed{x^2}$ = 2998.49
- $(-17)^2$ = 17 +/- $\boxed{x^2}$ = 289
- $(-12)^2$ = 12 +/- $\boxed{x^2}$ = 144

The square of a negative number is positive

Another look at the square of a number leads to the idea of the square root of a number. The square root of a number is a value that when multiplied by itself equals the original number.

Square Root of a Number:

Example:

- 4 = 2 x 2, the square of 2 = 4, $\sqrt{4}$ = 2, 2 is the square root of 4
- 9 = 3 x 3, the square of 3 = 9, $\sqrt{9}$ = 3, 3 is the square root of 9

Indices:

Power n (the number of times the multiplication is repeated)

a^n

Base a (the number that is multiplied repeatedly)

Indices take the form $base^{power}$. Roots are a way of getting back to the base, in the same way as divide reverses multiply.

Example:

- $3^2 = 9$, but $\sqrt{9}$ gets you back to your base 3
- $2^3 = 8$ but $\sqrt[3]{8}$ gets you back to your base 2

Complete the following exercise:

A. $8^2 =$	**B.** $\sqrt{40} =$
C. $9^2 =$	**D.** $\sqrt{60} =$
E. $16^2 =$	**F.** $\sqrt{336} =$
G. $26^2 =$	**H.** $\sqrt{416} =$
I. $18 \times 18 =$	**J.** $\sqrt{756} =$
K. $26 \times 21^2 =$	**L.** $\sqrt{(216 + 28)} =$

1.4 Mixed Operations

Some calculations involve a string of numbers and operation signs such as +, -, x, ÷

Example 1:

8 x 3 – 10 =

Which operation do we perform first? The calculator knows, it is programmed to know. This is part of algebraic logic design. The calculator will choose "x" over "-"(multiplication over subtraction). This means that multiplication has precedence over subtraction or that multiplication must always be done first.

8 x 3 = 24

24 – 10 = 14 (Answer)

Example 2:

In the example below, we are going to use a very importance rule in mathematics when carrying out a longer sum. The rule is called BEDMAS

B	E	D	M	A	S
Brackets	Exponents	Division	Multiplication	Addition	Subtraction
( )	e^2	÷	x	+	-

$5 \times (10 - 6)^2 \div 2 + 1$

$5 \times (4)^2 \div 2 + 1$ (Brackets)

$5 \times 16 \div 2 + 1$ (Exponents)

$=5 \times 8 + 1$ (Division)

$=40 + 1$ (Multiplication)

$=40 + 1$ (Addition)

$=41$ (Answer)

Complete the following exercise:

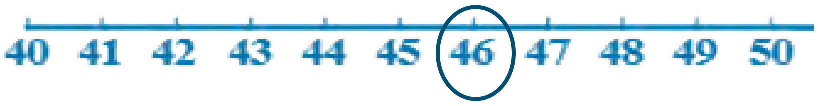
A. 5 x 6 – 8 =	**B.** (8 x 15) + 28 - 39 =
C. 8 – 4 x 2 =	**D.** (84 ÷ 14) + 8 -9 =
E. 16 ÷ 4 – 2 =	**F.** (20 x 6) ÷ 12 + 6 =
G. 8 x (12 ÷ 4) – 16 + 7 =	**H.** 5 x (16 – 8) ÷ 8 +12 =
I. 64 ÷ 4 x 8 – 2 =	**J.** 12 + 8 – 6 x (8 ÷ 2) =
K. $7 \times 9^2 + 45 =$	**L.** $88 - 1^2 + 1 =$
M. $3 \times 9^2 + 94 =$	**N.** $8 \times (7^2 - 5) =$

1.5 Rounding Off Numbers

When we round numbers, we lose accuracy, but the advantage is that we can do calculations more easily and quickly to make estimations.

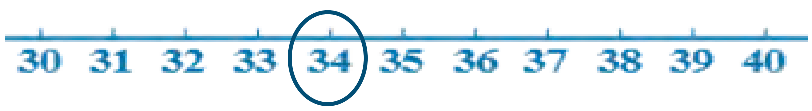
Rules for rounding to the nearest 10:

1. If the number we are rounding **ends** with 5, 6, 7, 8, or 9, we round the number up.
   Example:
   46 rounded to the nearest ten is 50



   67 rounded to the nearest ten is 70

2. If the number you are rounding **ends** with 0, 1, 2, 3, or 4, we round the number down.
   Example:
   34 rounded to the nearest ten is 30

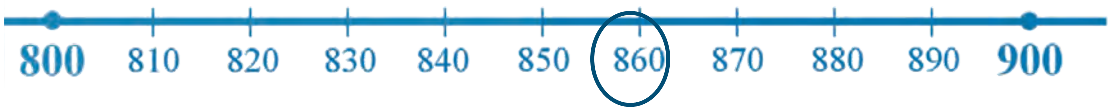


Round the following numbers to the nearest 10:

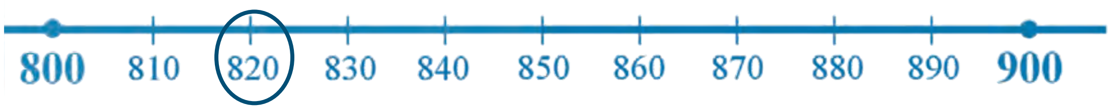
A. 97 =	**B.** 86 =
C. 18 =	**D.** 46 =
E. 28 =	**F.** 48 =
G. 11 =	**H.** 29 =
I. 36 =	**J.** 79 =

Rules for rounding to the nearest 100:

1. If the number we are rounding **ends** in 50 or more, we round the number up.
   Example:
   860 rounded to the nearest 100 is 900



2. If the number we are rounding **ends** in less than 50, we round the number down.
   Example:
   820 rounded to the nearest 100 is 800



Round the following numbers to the nearest 100:

A. 623 =	**B.** 765 =
C. 145 =	**D.** 238 =
E. 6,722 =	**F.** 5,864 =
G. 8,835 =	**H.** 2,327 =
I. 6,562 =	**J.** 5,449 =
K. 4,477 =	**L.** 4,317 =
M. 7,483 =	**N.** 5,698 =

1.6 Rounding Off Decimals

Rule:

- Count the digits after the decimal to the required number of decimal places
- Look at the number to the right of the number
    If this digit is 5 or more, we increase the previous digit by 1
    If it is less than 5, we leave the previous digit as it is
- Remove all the digits to the right

Example:

	1 Decimal Place	2 Decimal Places	3 Decimal Places
2.5643	2.5643 2.6	2.5643 2.56	2.5643 2.564

Complete the following exercise:

	1 Decimal Place	2 Decimal Places	3 Decimal Places
5.3742			
3.3787			
0.9834			
0.0768			
7.4367			
0.0813			
6.4874			
0.6677			
3.2473			
0.5368			
0.3466			
7.2364			

Complete the following questions:

Q1.

You need the following lengths of metal for an item you are making: 1.99m, 4.93m, 6.03m and 5.14m. You have a 20m length in the workshop. Will you have enough?

Note: Round each length to a whole number before adding.

Q2.

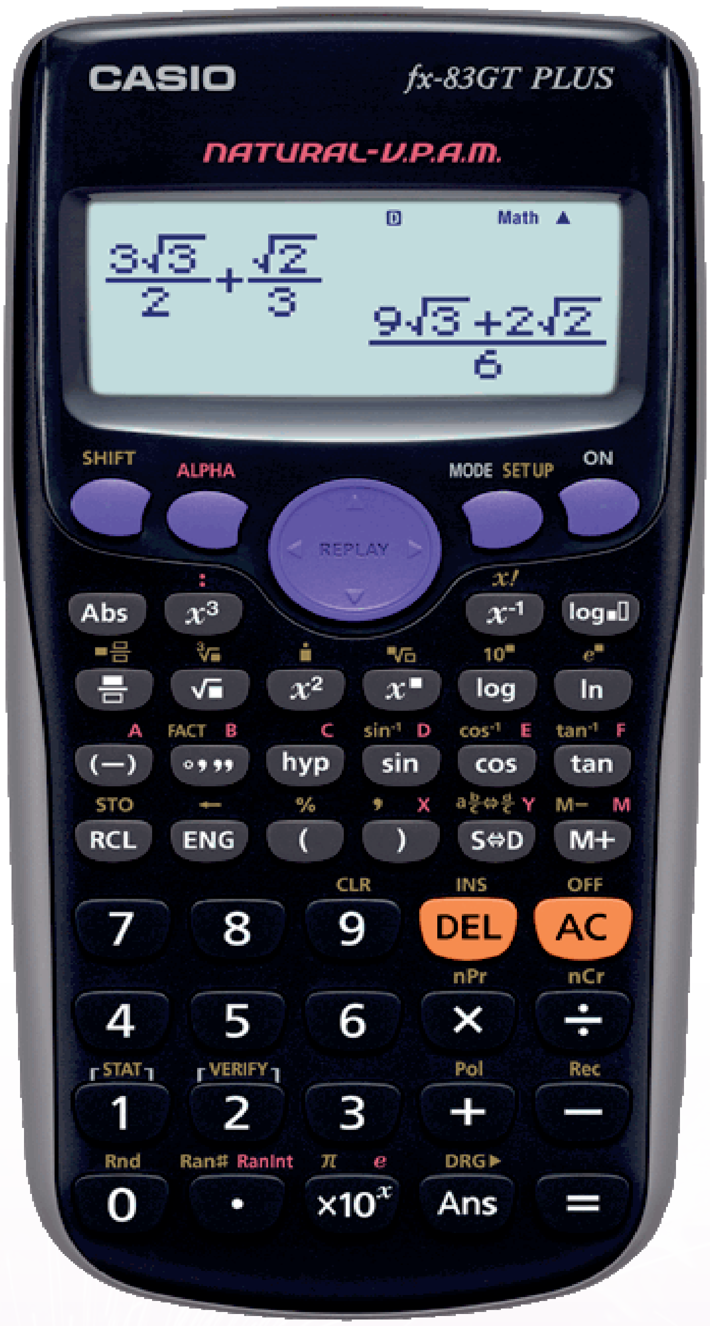
You have been quoted €795 for 4 sheets of metal. About how much is that per sheet?

Q3.

Estimate the value of:

$$\frac{9.02 + 6.95}{0.042 \text{ x } 11}$$

1.7 Using the Calculator



2nd Function

Many of the keys on the calculator can do 2 functions. Like the computer keyboard. To carry out the function in yellow over each key, you must first press the shift key.

To use the 2nd function, you press the shift key

SHIFT

e.g

% , you press shift and the bracket button: (

Getting Decimal Answers

Your calculator will often give the answer as a fraction.

To change it to a decimal number, press:

S⇔D

Brackets

For some calculations you will need to use brackets:

( )

Try typing the following calculation into your calculator:

(7 + 3) x 5

Check that you get 50.

Powers/Indices

To get a power of any number, use the following button:

$x^{\blacksquare}$

For example, to find 15^3, type:

1 5 $x^{\blacksquare}$ 3

You may also find these buttons useful for numbers squared:

x^2 x^3

Square Root To find the square root of any value: √■ For example, find $\sqrt{100}$	**Fractions** Your calculator has a button for entering fractions:  Note: You will need to use  to move to the second box.
Negative Numbers To enter negative numbers, use this: (-)	**Standard Form** You can enter numbers in standard form using this button: $x10^x$
Trigonometry Functions sin cos tan	**Inverse functions** Use the shift key then the trig function: SHIFT Sin-1 Cos- Tan-

*** All Calculators are different. The above description is for Casio calculators. It is important to be familiar with the functions of the calculator that you are using.

1.8 Arithmetic Review & Practical Application Questions

Q1.

A stockperson has just received a shipment of 126 cans of paint that need to be displayed. If each shelf holds 9 cans, how many shelves will he need to display all the cans of paint?

a. 9 shelves
b. 14 shelves
c. 16 shelves
d. 32 shelves

Q2.

A painter has a service charge of €35 per hour plus the cost of the paint. How much is the total bill for a project if she spent €32 on paint and worked for six hours?

a. €67
b. €192
c. €227
d. €242

Q3.

Victoria is trying to figure out how many cans of paint she will need for her next job. She has three rooms to paint. One room has 480 ft^2 to paint, and the other two rooms have 320 ft^2 each. If each one gallon can of paint covers 200 ft^2, what is the minimum number of cans that Victoria will have to buy?

a. 4 cans
b. 6 cans
c. 10 cans
d. 5 cans

Unit 2: Fractions

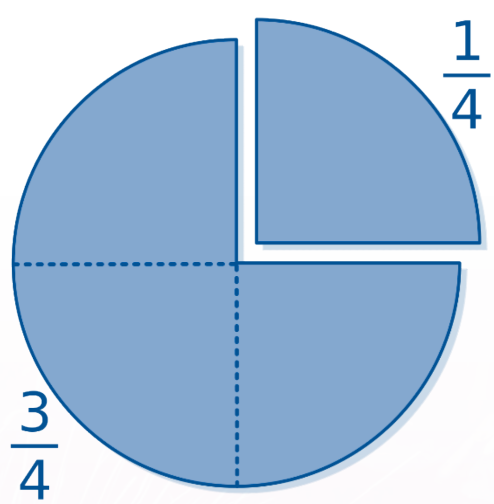
2.1 What is a Fraction?

A **fraction** simply tells us how many parts of a whole we have. You can recognize a fraction by the slash that is written between the two numbers. We have a top number, the **numerator**, and a bottom number, the **denominator**. For example, $\frac{3}{5}$ is a fraction

$$\frac{3}{5} \begin{array}{l} \longrightarrow \text{Numerator} \\ \longrightarrow \text{Denominator} \end{array}$$

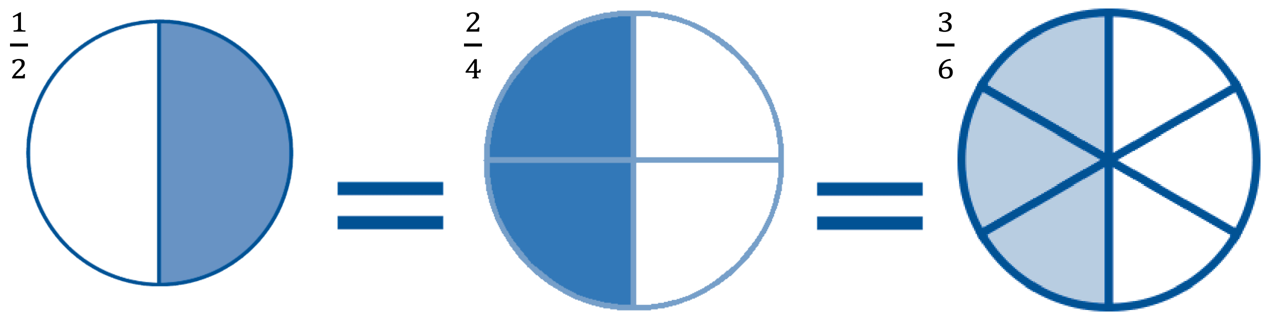
The numerator tells us how many parts we have, and the denominator tells us how many parts there were altogether.

E.g.: If a pizza has 4 pieces and we eat one, then there are 3 pieces of pizza left. We ate $\frac{1}{4}$ of the pizza, so there is and there are $\frac{3}{4}$ of the pizza left.



2.2 Equivalent Fractions

Equivalent fractions represent the same part of a whole. They look different but have the same overall value. Look at the fractions below. Half of each circle is shaded, but the fraction representing the value half is written in lots of different ways. We can keep breaking something into more and more parts giving us more equivalent



Complete the following exercise:

A. $\frac{1}{2} = \frac{}{4}$	**B.** $\frac{1}{3} = \frac{}{6}$
C. $\frac{1}{3} = \frac{}{15}$	**D.** $\frac{1}{4} = \frac{4}{}$
E. $\frac{1}{5} = \frac{}{25}$	**F.** $\frac{1}{3} = \frac{3}{}$
G. $\frac{1}{2} = \frac{4}{}$	**H.** $\frac{1}{5} = \frac{3}{}$

2.3 Simplifying Fractions

Simplifying (or *reducing*) fractions means to make the fraction as simple and as small as possible.

Example 1:

Simplify the fraction $\frac{8}{12}$

The largest number that goes exactly into both 8 and 12 is 4, so the Highest Common Factor (HCF) is 4.

Divide both top and bottom by 4:

$$\frac{8}{12} \xrightarrow{\div 4} \frac{2}{3}$$

The fraction simplifies to $\frac{2}{3}$

Example 2:

Simplify the fraction $\frac{10}{35}$

The largest number that goes exactly into both 10 and 35 is 5, so the Highest Common Factor (HCF) is 5.

Divide both top and bottom by 5

$$\frac{10}{35} \xrightarrow{\div 5} \frac{2}{7}$$

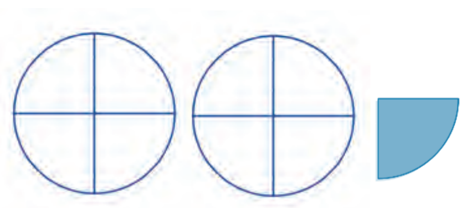
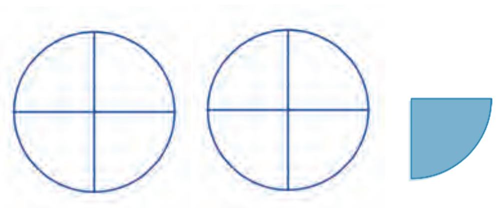
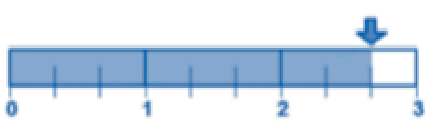
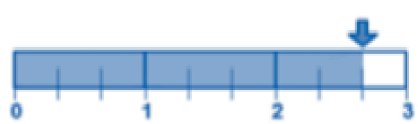
The fraction simplifies to $\frac{2}{7}$

Complete the following exercise:

A. $\frac{2}{4} =$	**B.** $\frac{3}{6} =$
C. $\frac{5}{15} =$	**D.** $\frac{6}{9} =$
E. $\frac{2}{8} =$	**F.** $\frac{3}{12} =$
G. $\frac{2}{10} =$	**H.** $\frac{3}{15} =$
I. $\frac{10}{25} =$	**J.** $\frac{3}{18} =$
K. $\frac{12}{20} =$	**L.** $\frac{24}{28} =$
M. $\frac{12}{32} =$	**N.** $\frac{30}{48} =$
O. $\frac{49}{56} =$	**P.** $\frac{33}{36} =$

2.4 Mixed & Improper Fractions

Earlier we described a fraction as being part (slice of pizza) of a full thing. These fractions that we have been looking at are called Proper Fractions. But when we have more than a whole, 2 and a bit or 3 and a bit or whatever, there are two ways of describing them mathematically; Mixed Fractions and Improper Fractions.

Mixed Fractions A whole number and a proper fraction	Improper Fractions Counts the parts
 Two whole pieces and one part = $2\frac{1}{4}$	 9 parts = $\frac{9}{4}$
 Two whole pieces, and 2 parts = $2\frac{2}{3}$	 8 parts = $\frac{8}{3}$

Examples:

- Convert $\frac{8}{6}$ to a mixed fraction = $1\frac{2}{6}$ (8 ÷ 6 = 1 and 2 left over)
- Convert $\frac{13}{6}$ to a mixed fraction = $2\frac{1}{6}$ (13 ÷ 6 = 2 and 1 left over)

- Convert $2\frac{1}{4}$ to an improper fraction:
  - Multiply the whole number by the denominator
    2 x 4 = 8
  - Add this figure to the numerator 8 + 1 = 9, 9 parts in total
  - Answer = $\frac{9}{4}$

- Convert $3\frac{7}{2}$ to an improper fraction:
  - Multiply the whole number by the denominator
    3 x 2 = 6
  - Add this figure to the numerator 6+ 1 = 7, 7 parts in total
  - Answer = $\frac{7}{2}$

- Convert the following improper fractions to mixed fractions:

A. $\frac{7}{5} =$	**B.** $\frac{9}{4} =$
C. $\frac{5}{3} =$	**D.** $\frac{22}{9} =$
E. $\frac{13}{7} =$	**F.** $\frac{9}{2} =$
G. $\frac{7}{3} =$	**H.** $\frac{17}{3} =$

Convert the following Mixed Fractions to Improper Fractions:

A. $2\frac{1}{4} =$	**B.** $8\frac{3}{8} =$
C. $5\frac{1}{3} =$	**D.** $2\frac{5}{6} =$
E. $3\frac{1}{2} =$	**F.** $4\frac{1}{2} =$
G. $10\frac{7}{12} =$	**H.** $9\frac{1}{4} =$
I. $7\frac{5}{6} =$	**J.** $10\frac{3}{7} =$
K. $3\frac{1}{4} =$	**L.** $2\frac{2}{6} =$
M. $2\frac{2}{6} =$	**N.** $3\frac{3}{6} =$

2.5 Addition of Fractions

- Make sure the bottom numbers (the denominators) are the same
- Add the top numbers (the numerators), put the answer over the denominator
- Simplify the fraction (if needed)

Example 1:

$$\frac{1}{4} + \frac{1}{4} = \frac{2}{4} = \frac{1}{2}$$

Example 2:

$$\frac{3}{4} + \frac{3}{4} = \frac{6}{4} = 1\frac{2}{4} = 1\frac{1}{2}$$ (improper fraction to a mixed fraction)

Example 3:

$$\frac{1}{3} + \frac{1}{6}$$ We need a common denominator (6)

$$\frac{(1\,x\,2) + (1\,x\,1)}{6} = \frac{2+1}{6} = \frac{3}{6} = \frac{1}{2}$$

Example 4:

$$\frac{1}{4} + \frac{2}{3}$$ We need a common denominator (12)

$$\frac{(1\,x\,3) + (2\,x\,4)}{12} = \frac{3+8}{12} = \frac{11}{12}$$

Add the following fractions with like denominators:

A. $\frac{1}{4} + \frac{1}{4} =$	**B.** $\frac{1}{6} + \frac{2}{6} =$
C. $\frac{1}{4} + \frac{2}{4} =$	**D.** $\frac{1}{3} + \frac{1}{3} =$
E. $\frac{2}{4} + \frac{3}{4} =$	**F.** $\frac{1}{8} + \frac{3}{8} =$
G. $\frac{1}{5} + \frac{2}{5} =$	**H.** $\frac{1}{7} + \frac{5}{7} =$

Add the following fractions with unlike denominators:

A. $\frac{1}{4} + \frac{1}{3} =$	**B.** $\frac{1}{6} + \frac{2}{3} =$
C. $\frac{1}{8} + \frac{2}{4} =$	**D.** $\frac{1}{4} + \frac{1}{2} =$
E. $\frac{2}{5} + \frac{1}{4} =$	**F.** $\frac{3}{4} + \frac{1}{16} =$
G. $\frac{1}{4} + \frac{3}{8} =$	**H.** $\frac{2}{5} + \frac{1}{10} =$

2.6 Subtraction of Fractions

- Make sure the bottom numbers (the denominators) are the same
- Subtract the top numbers (the numerators), put the answer over the denominator
- Simplify the fraction (if needed)

Example 1:

$\frac{2}{4} - \frac{1}{4} = \frac{1}{4}$

Example 2:

$\frac{3}{4} - \frac{1}{4} = \frac{2}{4} = \frac{1}{2}$

Example 3:

$\frac{1}{3} - \frac{1}{6}$ We need a common denominator (6)

$$\frac{(1\ x\ 2) - (1\ x\ 1)}{6} = \frac{2-1}{6} = \frac{1}{6}$$

Example 4:

$\frac{3}{4} - \frac{2}{3}$ We need a common denominator (12)

$$\frac{(3\ x\ 3) - (2\ x\ 4)}{12} = \frac{9-8}{12} = \frac{1}{12}$$

Subtract the following fractions with like denominators:

A. $\frac{2}{4} - \frac{1}{4} =$	**B.** $\frac{3}{6} - \frac{2}{6} =$
C. $\frac{8}{10} - \frac{2}{10} =$	**D.** $\frac{10}{12} - \frac{3}{12} =$
E. $\frac{7}{9} - \frac{5}{9} =$	**F.** $\frac{4}{8} - \frac{1}{8} =$
G. $\frac{7}{8} - \frac{2}{8} =$	**H.** $\frac{6}{7} - \frac{5}{7} =$

Subtract the following fractions with unlike denominators:

A. $\frac{2}{4} - \frac{1}{3} =$	**B.** $\frac{2}{9} - \frac{1}{6} =$
C. $\frac{5}{7} - \frac{1}{3} =$	**D.** $\frac{5}{7} - \frac{2}{5} =$
E. $\frac{5}{9} - \frac{1}{3} =$	**F.** $\frac{6}{10} - \frac{1}{8} =$
G. $\frac{9}{16} - \frac{1}{4} =$	**H.** $\frac{2}{5} - \frac{1}{10} =$

2.7 Multiplying Fractions

- Multiply the numerators of the fractions to get the new numerator
- Multiply the denominators of the fractions to get the new denominator
- Simplify the resulting fraction if possible

Example 1:

$\frac{2}{4} \text{ x } \frac{1}{4} = \frac{2}{16} = \frac{1}{8}$

Example 2:

$\frac{3}{4} \text{ x } \frac{1}{2} = \frac{3}{8}$

Complete the following exercise:

A. $\frac{2}{4} \text{ x } \frac{3}{4} =$	**B.** $\frac{5}{6} \text{ x } \frac{1}{2} =$
C. $\frac{7}{9} \text{ x } \frac{1}{12} =$	**D.** $\frac{1}{2} \text{ x } \frac{1}{3} =$
E. $\frac{4}{9} \text{ x } \frac{2}{3} =$	**F.** $\frac{5}{10} \text{ x } \frac{1}{2} =$
G. $\frac{1}{8} \text{ x } \frac{2}{4} =$	**H.** $\frac{3}{5} \text{ x } \frac{3}{7} =$

2.8 Dividing Fractions

- You turn the second fraction upside down (reciprocal)
- Multiply out the fractions
- Simplify the resulting fraction if possible

Example 1:

$$\frac{2}{4} \div \frac{1}{4} = \quad \frac{2}{4} \times \frac{4}{1} \quad = \frac{8}{4} \quad = 2$$

(Turn the second fraction upside down and multiply)

Example 2:

$$\frac{5}{10} \div \frac{2}{4} = \quad \frac{5}{10} \times \frac{4}{2} \quad = \frac{20}{20} \quad = 1$$

(Turn the second fraction upside down and multiply)

Complete the following exercise:

A. $\frac{7}{10} \div \frac{2}{4} =$	**B.** $\frac{1}{3} \div \frac{5}{10} =$
C. $\frac{1}{2} \div \frac{2}{5} =$	**D.** $\frac{6}{10} \div \frac{1}{3} =$
E. $\frac{2}{10} \div \frac{1}{5} =$	**F.** $\frac{2}{5} \div \frac{1}{3} =$
G. $\frac{1}{5} \div \frac{2}{4} =$	**H.** $\frac{4}{10} \div \frac{2}{4} =$

2.9 Finding a fraction of a number

- You divide the number by the bottom (the denominator)
- You then multiply this figure by the top (the numerator

Example:

Find $\frac{2}{5}$ of 200

200 ÷ 5 = 40 (divide by the denominator)

40 x 2 = 80 (multiply by the numerator)

Answer = 80

Complete the following exercise:

A. Find $\frac{1}{4}$ of 200	**B.** Find $\frac{4}{5}$ of 400
C. Find $\frac{3}{4}$ of 600	**D.** Find $\frac{5}{6}$ of 840
E. Find $\frac{1}{3}$ of 540	**F.** Find $\frac{1}{8}$ of 1200
G. Find $\frac{2}{3}$ of 600	**H.** Find $\frac{3}{5}$ of 60

Unit 3: Decimals & Percentages

3.1 Converting Fractions to Decimals

- To convert a fraction to a decimal, you divide the top (numerator) by the bottom (denominator)

Example 1:

$\frac{1}{5} = 1 \div 5 = 0.2$

Example 2:

$\frac{1}{4} = 1 \div 4 = 0.25$

Convert the following fractions to decimals:

A. $\frac{2}{4} =$	**B.** $\frac{3}{6} =$
C. $\frac{1}{3} =$	**D.** $\frac{3}{5} =$
E. $\frac{7}{10} =$	**F.** $\frac{3}{12} =$
G. $\frac{2}{10} =$	**H.** $\frac{5}{2} =$
I. $\frac{7}{4} =$	**J.** $\frac{10}{3} =$

3.2 Converting Decimals to Fractions

One Decimal Place:

- If there is one place after the decimal, then you put the number over 10
- Simplify the fraction if needed

Example:

$$0.6 = \frac{6}{10} = \frac{3}{5}$$

Two Decimal Places:

- If there are two places after the decimal, then you put the number over 100
- Simplify the fraction if needed

Example:

$$0.25 = \frac{25}{100} = \frac{1}{4}$$

Three Decimal Places:

- If there are three places after the decimal, then you put the number over 1000
- Simplify the fraction if needed

Example:

$$0.625 = \frac{625}{1000} = \frac{5}{8}$$

Convert the following decimals to fractions:

A. 0.32 =	**B.** 1.1 =
C. 0.8 =	**D.** 2.3 =
E. 0.125 =	**F.** 3.8 =
G. 0.2 =	**H.** 2.45 =
I. 0.4 =	**J.** 3.24 =
K. 4.25 =	**L.** 1.8 =

3.3 Expressing one quantity as a percentage of another

- Write both quantities in the same units and remove the units
- Put the first number over the second number to make a fraction
- Multiply the fraction by 100

Example:

Express 80c as a percentage of €2.40

$\frac{80}{240}$ x 100 = $33\frac{1}{3}$ % or 33.33%

In each of the following, express the first quantity as a percentage of the second:

A. 15, 20	**B.** 300m, 1km
C. 20, 50	**D.** 4mm, 2cm
E. 3, 5	**F.** 900g, 2kg
G. 36, 300	**H.** 80cm, 2m
I. 60, 300	**J.** 20, 8

Complete the following questions:

Q1.

A student gets 142 marks out of 200 marks in an exam. What percentage is this?

Q2.

Which is the better record; 20 wins in 25, or 30 wins in 40?

Q3.

An auctioneer received €4100 as commission when she sold a house for €164,000. What percentage commission did the auctioneer receive?

Q4.

What percentage of €5.00 is 42c?

Q5.

A maths test is marked out of 40 marks. A student scores 28 marks. What percentage score did the student get?

3.4 Percentage Increase/Decrease

We use the following formula when we want to find out the percentage increase or decrease between two figures, amounts or quantities.

$$\frac{\text{Difference}}{\text{Original}} \text{ x } 100$$

Example:

A man's salary went from €40,000 to €44,800. Calculate the percentage increase in salary.

Difference = €44,800 – €40,000 = €4,800

$\frac{4,800}{40,000}$ x 100 = 12%

Q1.
A maths book had 350 pages. A new edition was published with 343 pages. Calculate the percentage decrease in the number of pages in the book.

Q2.
A man's salary increased from €30,900 to €33,063. Calculate the percentage increase in his salary?

Q3.
A student failed an exam obtaining a result of 45 marks. On repeating the exam, the student obtained 65 marks. What was the percentage increase in marks?

3.5 Finding a Given Percentage of a Quantity

- Divide the quantity by 100
- Multiply by the percentage asked for

Example:

25% of 200

200 ÷ 100 = 2

2 x 25 = 50

or

You can use the % button on your calculator

200 x 25 $\boxed{\%}$ = 50

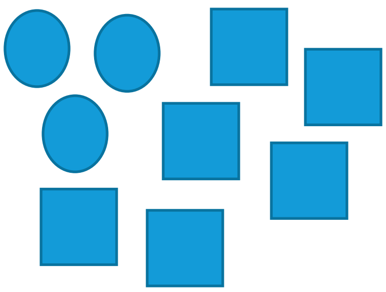
Complete the following exercise:

A. Find 10% of €380	**B.** Find 20% of €150
C. Find 40% of 180m	**D.** Find 12.5% of 400mm
E. Find 13% of 2,500m	**F.** Find 8% of €650
G. Find 7.5% of €6,500	**H.** Find 25% of 300m

Unit 4: Ratios

A ratio between two or more quantities is a way of measuring their sizes compared to each other. Ratios can be written in three different ways: the word "to", with a colon (:), or as a fraction.

Example:



- ratio of squares to circles is 3 to 6
- ratio of squares to circles is 3:6
- ratio of squares to circles is $\frac{3}{6}$

4.1 Expressing Ratios in their Simplest Form

We can reduce ratios in the same way we reduce fractions, by dividing by the highest common factor.

Examples:

12:15

4:5 (Divide each part by 3)

14:28:35

2:4:5 (Divide each part by 7)

0.25:0.75

1:3 (Divide each part by 0.25)

Express each of the following in its simplest form:

A. 4:8	**B.** 15:12
C. 15:20	**D.** 50:30
E. 3:9	**F.** 20:100
G. 12:30	**H.** 88:77
I. 6:10	**J.** 8:12:20
K. 14:21	**L.** 15:20:25
M. 20:200	**N.** 12:180:240

4.2 Proportional Parts

Ratios can be used to divide or share quantities.

To divide or share quantities:

- Add the ratios to get the total number of parts
- Divide the quantity by the total of the parts (this gives one part)
- Multiply this separately by each ratio

Example 1:

Divide €28 in the ratio 2:5

2 + 5 = 7 (7 parts in total)

28 ÷ 7 = 4 (1 part)

(2 x 4): (5 x 4)

8: 20

€8: €20

Example 2:

Divide 300Kg in the ratio 2:5:8

2 + 5 + 8 = 15 (15 parts in total)

300 ÷ 15 = 20 (1 part)

(20 x 2): (20 x 5): (20 x 8)

40: 100: 160

40kg: 100kg: 160kg

Complete the following questions:

Q1.
An alloy consists of copper, zinc and tin in the ratio 7:4:3 (by weight). Find the weight of each metal in 84kg of alloy.

Q2.
600g of alloy is made up of copper, tin and lead in the ratio 3:4:3 (by weight). How many grams of each metal are there in the alloy?

Q3.
The lengths of two pieces of metal are in the ratio of 5:2. The larger piece is 250mm. Find the length of the shorter piece.

Q4.
Copper and Zinc are mixed in the ratio of 19:6. The amount of copper used is 133kg. How many kg of zinc are used?

Unit 5: Units of Measurement

5.1 Measuring Length

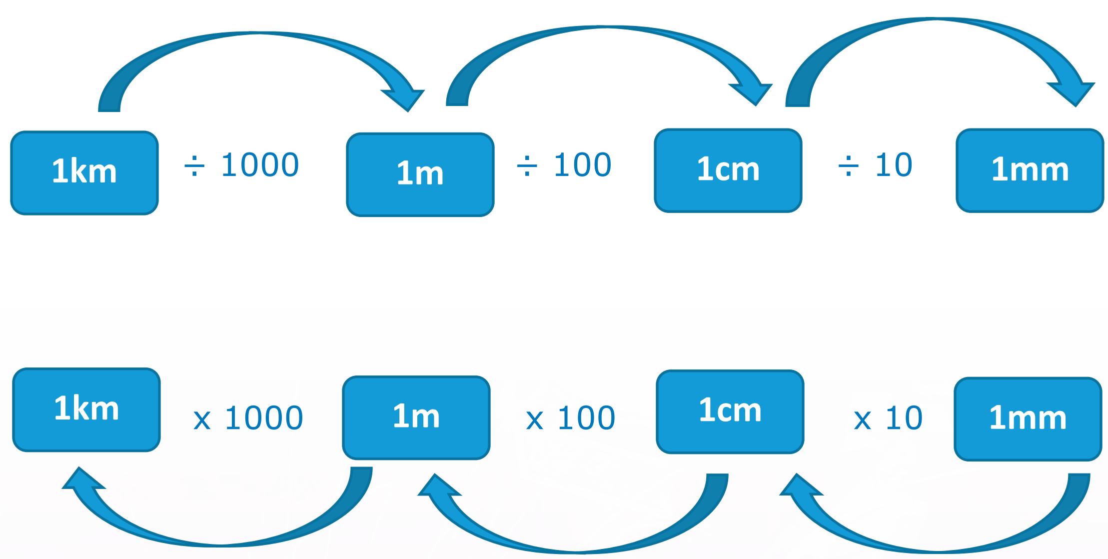
In the metal fabrication industry, the unit of measurement used is the metric unit. However, you may still come across imperial measures sometimes, so a knowledge of both is needed.

The metric units used are also known as **SI units**. The SI base units are the standard units of measurement defined by the International System of Units

Base units:

- The base unit of length is the metre (m)
- Its multiple unit is the kilometre (km) which is 1000 times larger:
  1m x 1000 = 1km
- The sub-unit is the milimetre (mm) which is 1000 times smaller:
  1m / 1000 = 1mm

Although the metre and the millimetre are the main units of length used, sometimes centimetres (cm) are encountered. Look at the conversions below:



5.2 Measuring Volume/Capacity

Volume

Volume is the amount of space occupied by a solid. Solids not only have length and width; they also have depth.
The space that solids take up cannot be measured in simple two-dimensional squares. Instead, solids need to be measured in terms of units which are three-dimensional. Units commonly used to measure the volume of objects include cubic millimetres, mm^3 and cubic centimetres, cm^3.

Capacity

Capacity is the amount that a container can hold. We can measure the capacity of a container by filling it up with cups of water and then counting the number of cups needed to fill it to the top.
To ensure that all measurements are the same, there are formal units to measure volume and capacity. These are litres, millilitres, and cubic centimetres.

Litres

Almost every liquid that we use is measured in litres (L).

Millilitres

The millilitre (mL) is used to measure smaller quantities of liquid. There are 1000 mL in 1 litre.

Useful conversions

1L = 1000mL
½ L = 500mL
¼ L = 250mL

Cubic centimetre

The cubic centimetre (cm^3) is a unit of volume measurement. The '3' indicates that three dimensions (length, width and depth) are used to find volume.

**1 cm^3 = .001 litres 91,000 cm^3 = 91 litres ** (Divide by 1000)

**1m^3 = 1000 litres 9m^3 = 9,000 litres **(Multiply by 1000)

Unit 6: Area & Volume

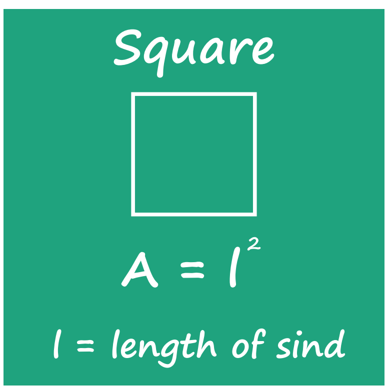
Area: The area of a shape is the size of a surface.

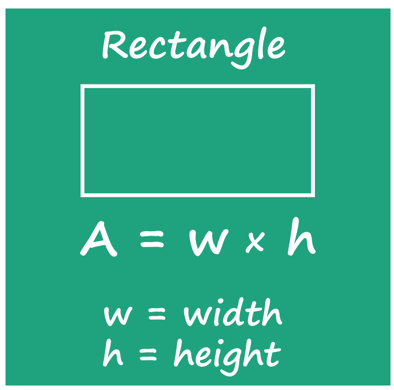
Perimeter: The perimeter of a shape is the distance around the edge of the shape.

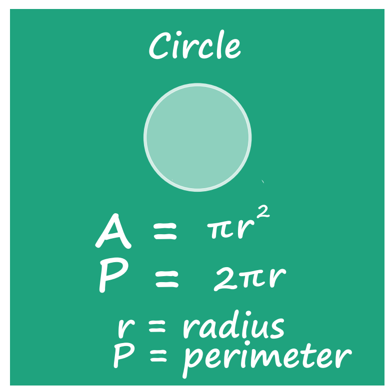
Volume: The amount of space/liquid/material a shape holds.

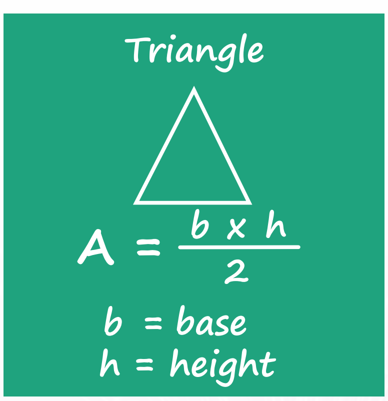
*Note: the value for π can be taken as 3.14 or 3.142

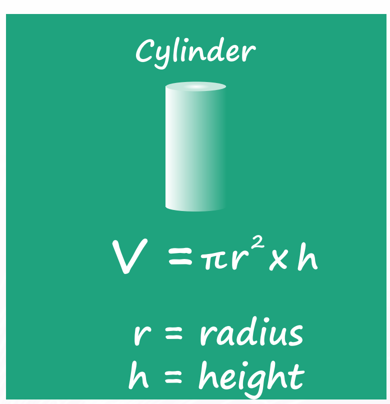
Formulae for this unit:

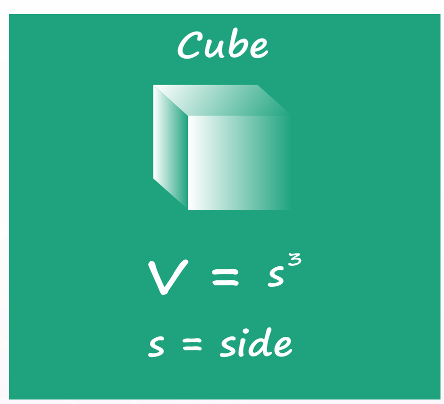






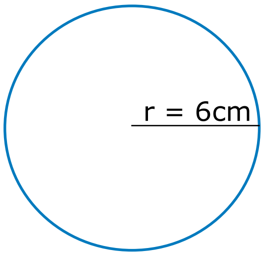






6.1 Find the area of the following shapes.

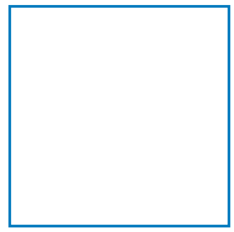
A.



F* =
A =

B.

L= 5cm



F =
A =

C.

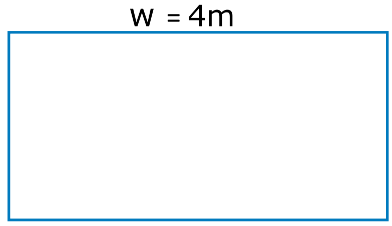
w = 7cm

h= 3cm

F =
A =

D.

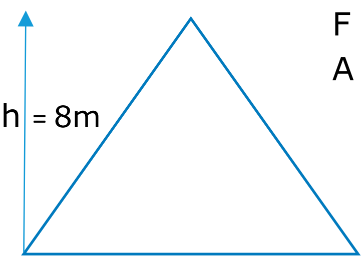
w = 4m



h = 2cm

F =
A =

E.

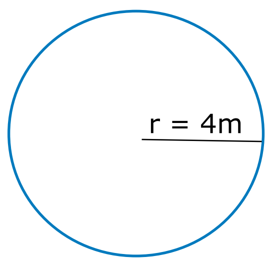


h = 8m

b = 10m

F =
A =

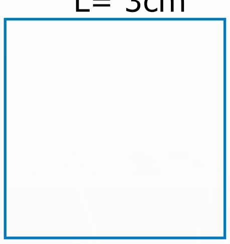
F.



F =
A =

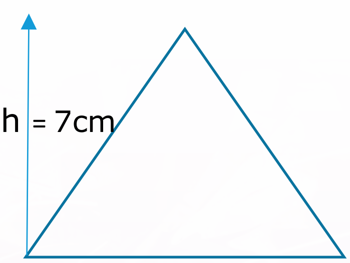
G.

L= 3cm



F =
A =

H.



h = 7cm

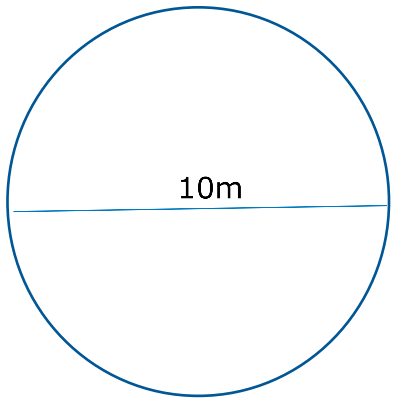
b = 9cm

F =
A =

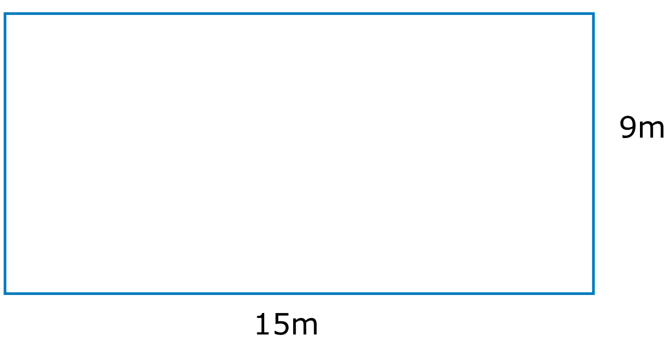
* F = Formula

Find the area and perimeter of the following:

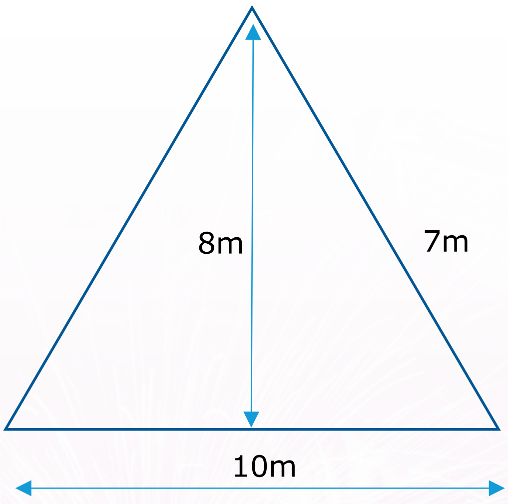
Q1.



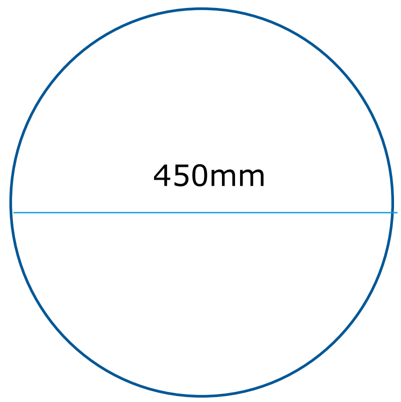
Q2.



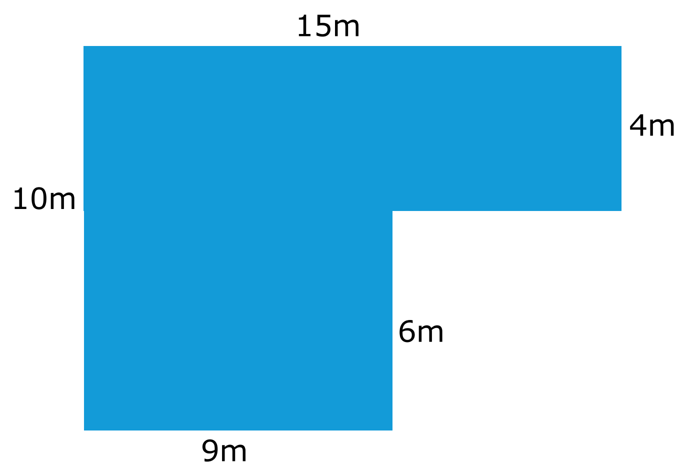
Q3.



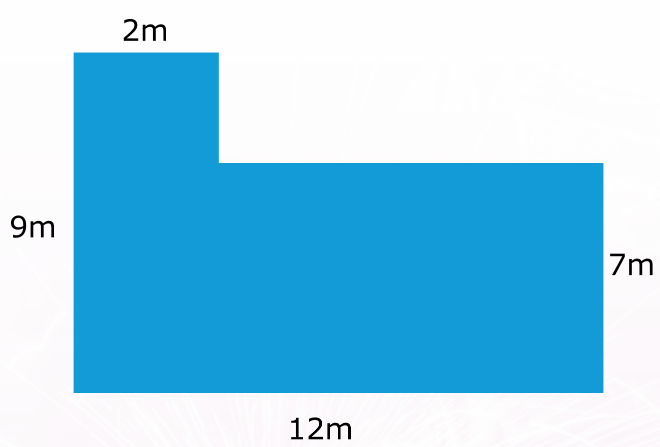
Q4.



Q5.



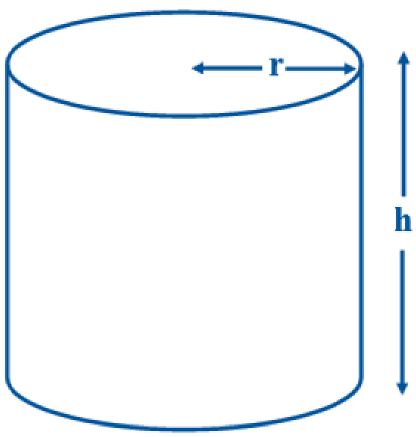
Q6.



6.2 Find the Volume of the Following Cylinders

Q1.

Use 3.142 for π. Volume = V =$\pi r^2 h$. Give your answer in litres. ($1m^3$=1000L)

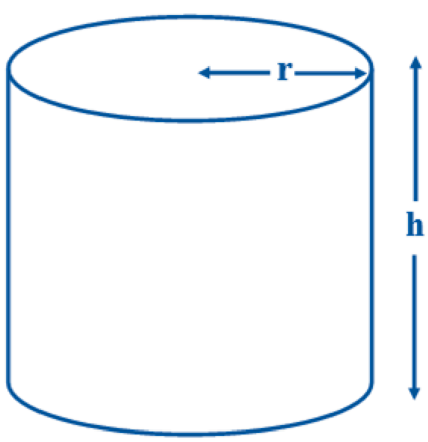


Radius = 6m

Height = 15m

Q2.

Use 3.142 for π. Volume = V =$\pi r^2 h$. Give your answer in litres. ($1m^3$=1000L)



Radius = 6cm

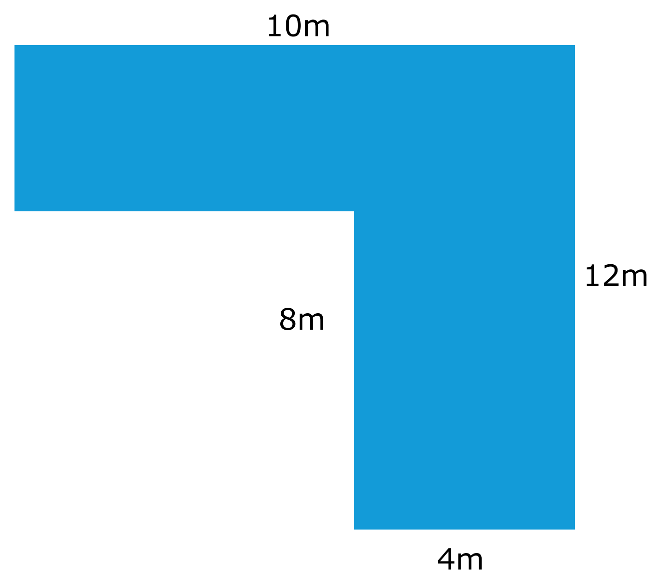
Height = 8cm

6.3 Area & Volume Review & Practical Application

Q1.

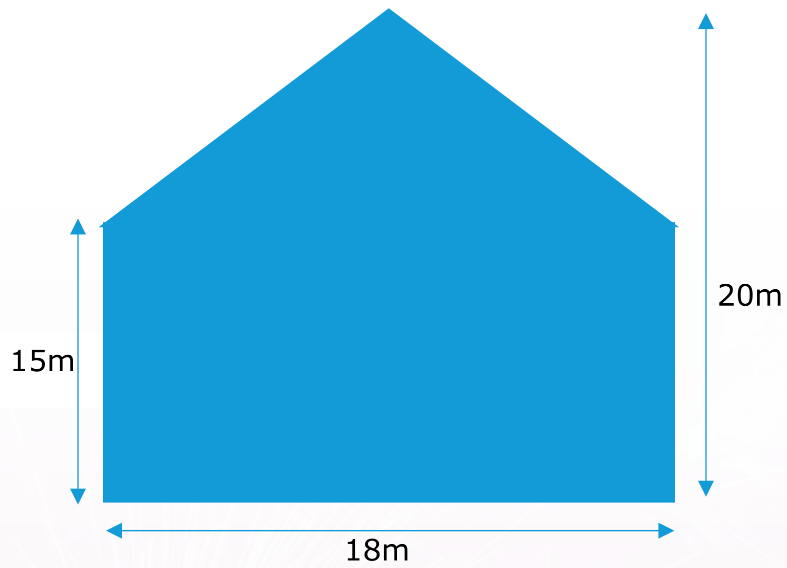
Look at the shape below:

(a) Find the area.
(b) Find the perimeter.



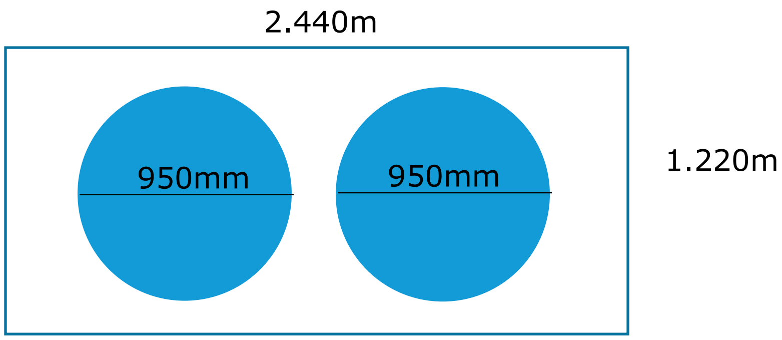
Q2.

Find the area of the gable end of the house below:



Q3.

2 circular discs are cut from a sheet of metal



(i) Calculate the area of the sheet of metal.

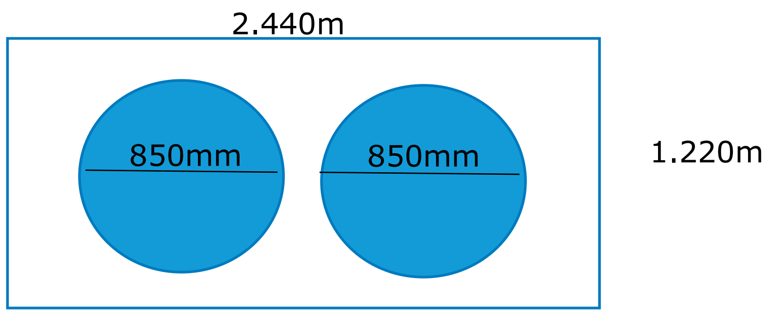
(ii) Calculate the area of 1 disc. (Convert the mm to m)

(iii) Calculate the area of 2 discs.

(iv) Calculate % area of wastage.

Q4.

3 circular discs are cut from a sheet of metal



(i) Calculate the area of the sheet of metal.

(ii) Calculate the area of 1 disc.

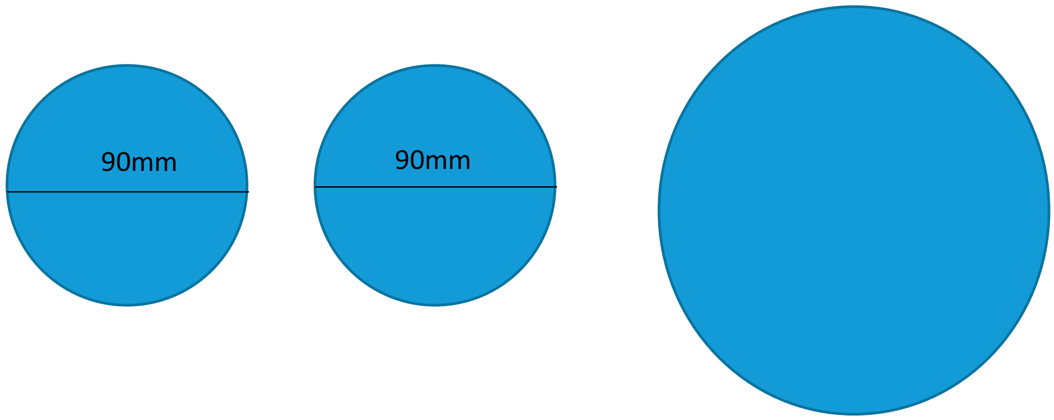
(iii) Calculate the area of 2 discs.

(iv) Calculate % area of wastage.

Q5.

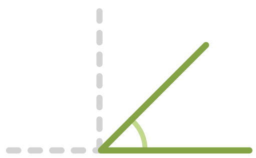
Two 90mm (inside diameter) water pipelines are to be replaced by one single pipeline.

Calculate the minimum diameter the new pipe should be in order to carry the same volume of water.

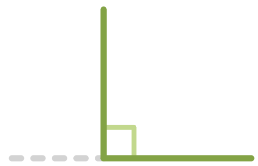


Unit 7: Angles

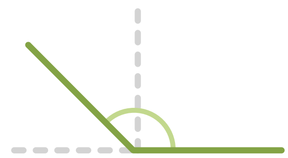
7.1 Classifying Angles



acute angle
less than 90°



right angle
90°

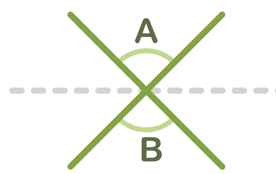


obtuse angle
between
90° and 180°

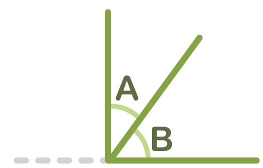
straight angle
180°

reflex angle
between
180° and 360°

a revolution
360°

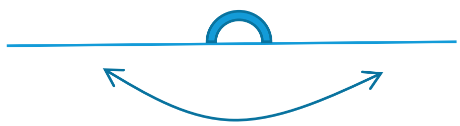
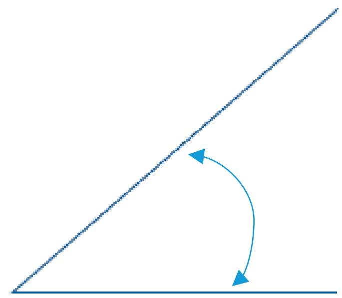
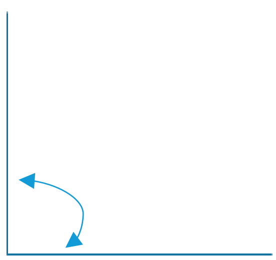
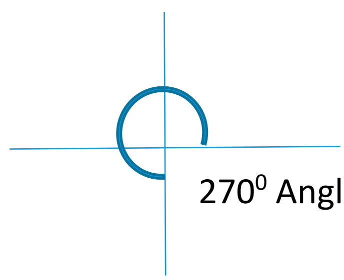
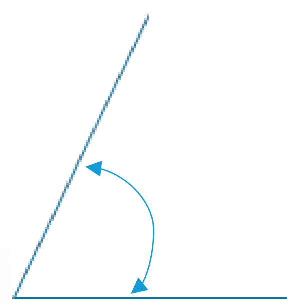


opposite angle
A = B



complementary angle
A + B = 90°

Classify each of the following angles:

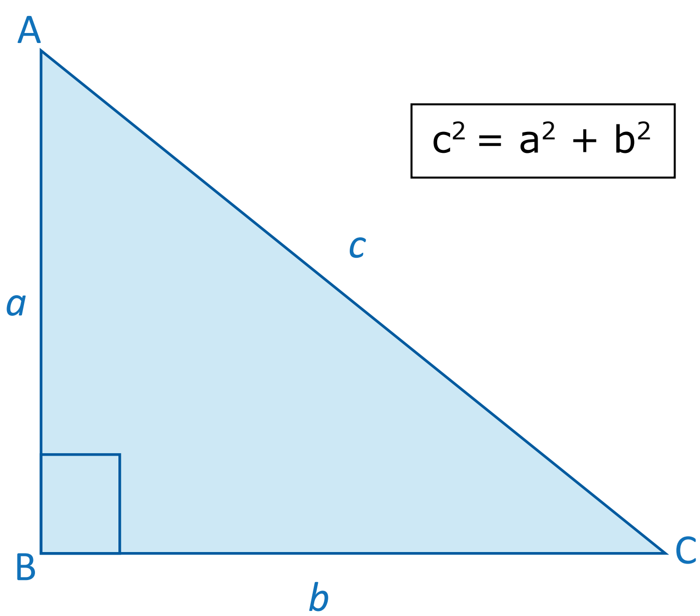
A.  180^0 Angle	**B.**  45° Angle
C.  90^0 Angle	**D.**  270^0 Angle
E.  135^0 Angle	**F.**  60^0 Angle

Unit 8: Pythagoras Theorem & Trigonometry

8.1 Pythagoras Theorem

The Pythagorean Theorem states that in a right-angled triangle, the square of the hypotenuse side is equal to the sum of the squares of the other two sides of the triangle. The hypotenuse is the longest side of a right-angled triangle. It is the side opposite the right angle. In the diagram below the hypotenuse is labelled C.

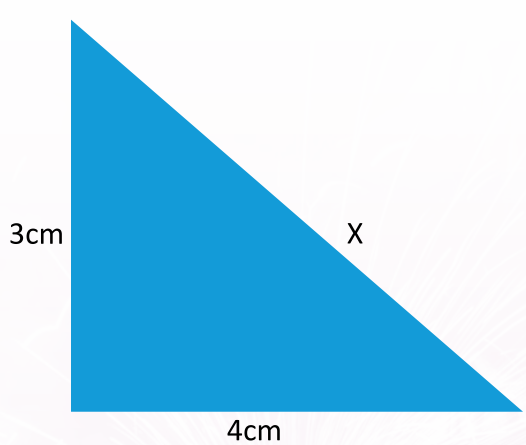
(BBC Bitesize: Pythagoras Theorem)



$c^2 = a^2 + b^2$

Example:

Find the side labelled x



$c^2 = a^2 + b^2$

$x^2 = 3^2 + 4^2$

$x^2 = 9 + 16$

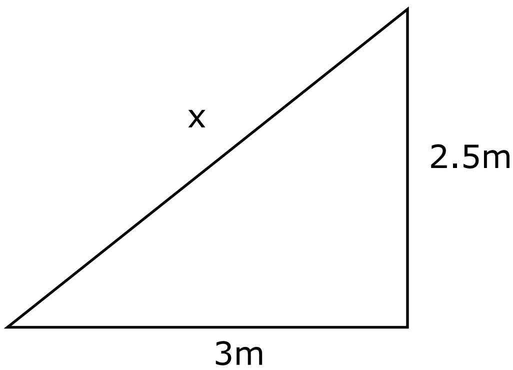
$x^2 = 25$

$x = \sqrt{25}$

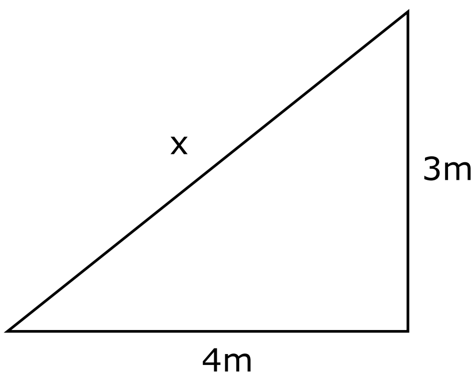
$x = 5$

Find the length of the side marked x in the following questions:

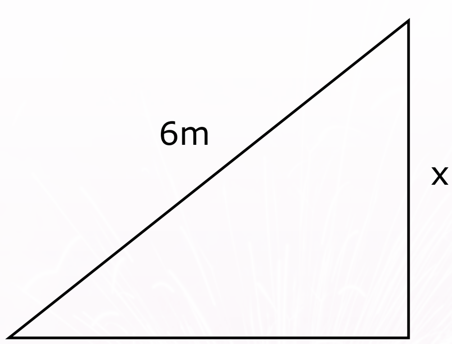
Q1.



Q2.

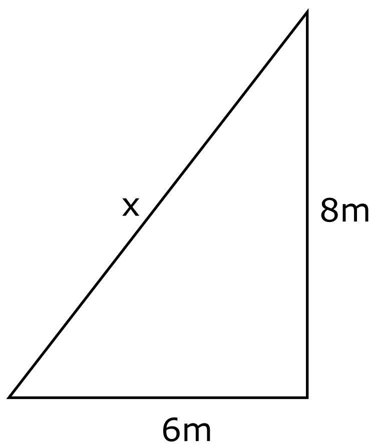


Q3.

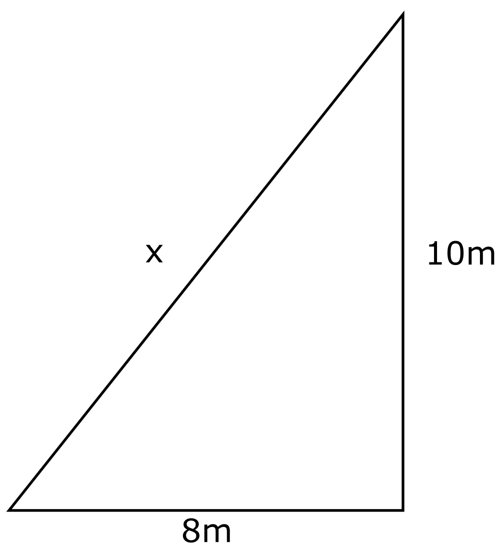


4m

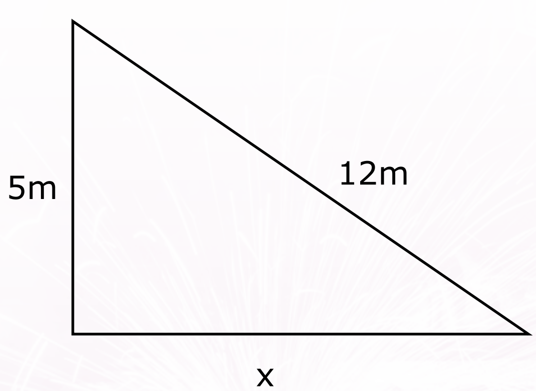
Q4.



Q5.



Q6.



8.2 Trigonometry

As a way of remembering how to compute the sine, cosine and tangent of an angle, you can use the following rhyme (or one you have learned):

Silly Old Harry, Caught A Herring, Trawling Off America

SOHCAHTOA

SOH - stands for Sine equals Opposite over Hypotenuse.

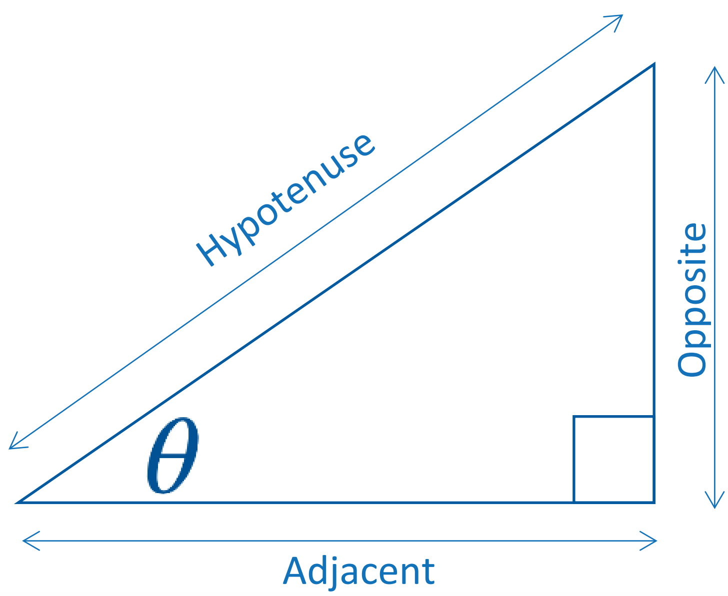
CAH - stands for Cosine equals Adjacent over Hypotenuse.

TOA - stands for Tangent equals Opposite over Adjacent.

$$\sin \emptyset = \frac{opposite}{hypotenuse}$$

$$\cos \emptyset = \frac{adjacent}{hyptenuse}$$

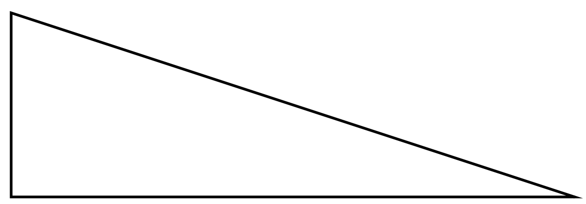
$$\tan \emptyset = \frac{opposite}{adjacent}$$



Complete the following questions:

Q1.

Label the sides of the triangle below



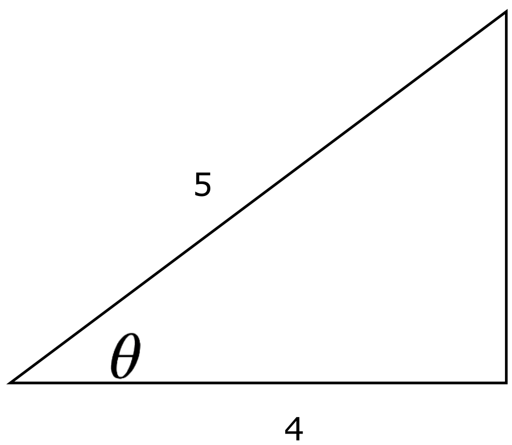
Sin =

Cos =

Tan =

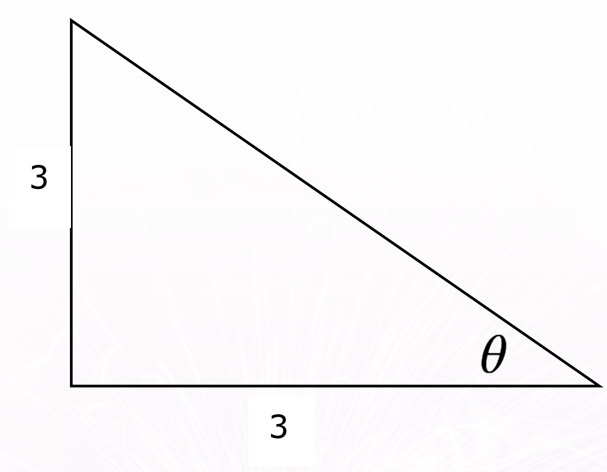
Q2.

Find the size of the angle shown:



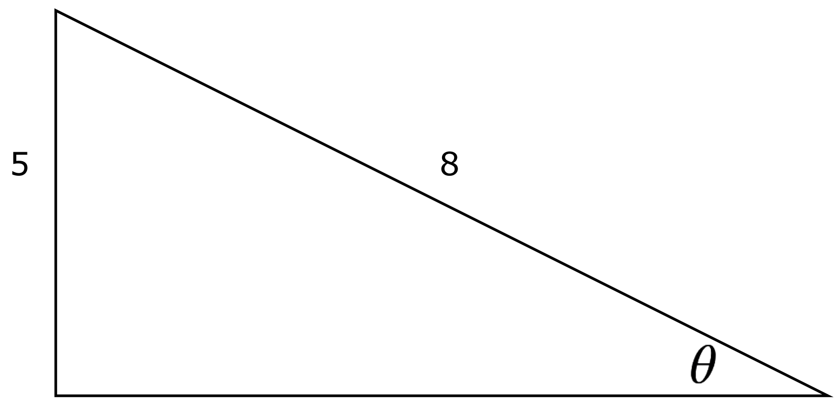
Q3.

Find the measure of the angle shown:



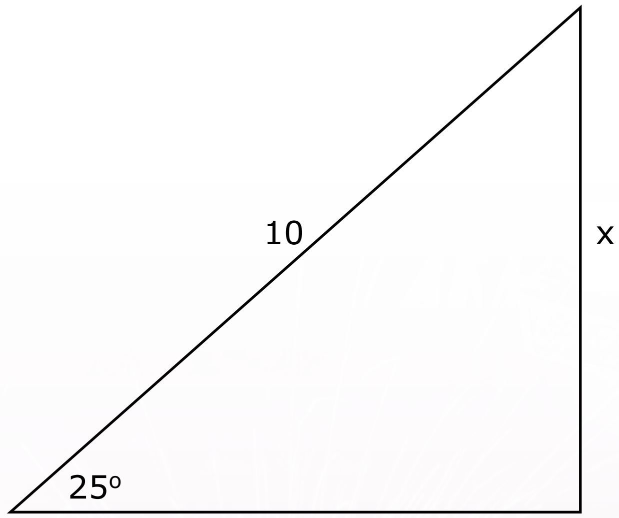
Q4.

Find the measure of the angle shown:



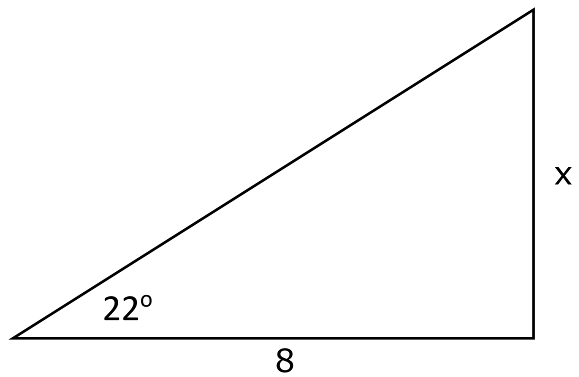
Q5.

Find the length of the side labelled x:



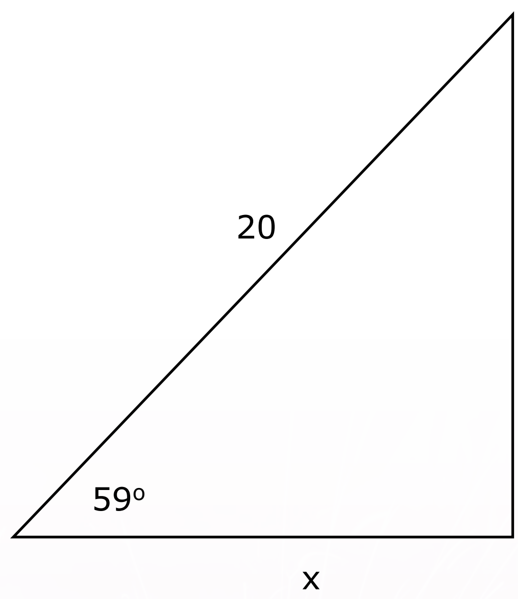
Q6.

Find the length of the side labelled x:



Q7.

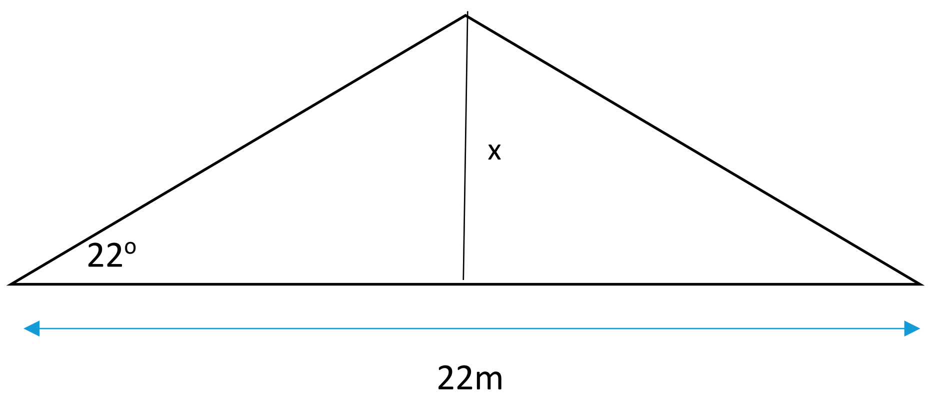
Find the length of the side labelled x:



8.3 Pythagoras Theorem & Trigonometry Practical Application

Q1.

A roof truss spans 22m and is angled at the shoe connection to 26º. Calculate the ridge height of the truss.



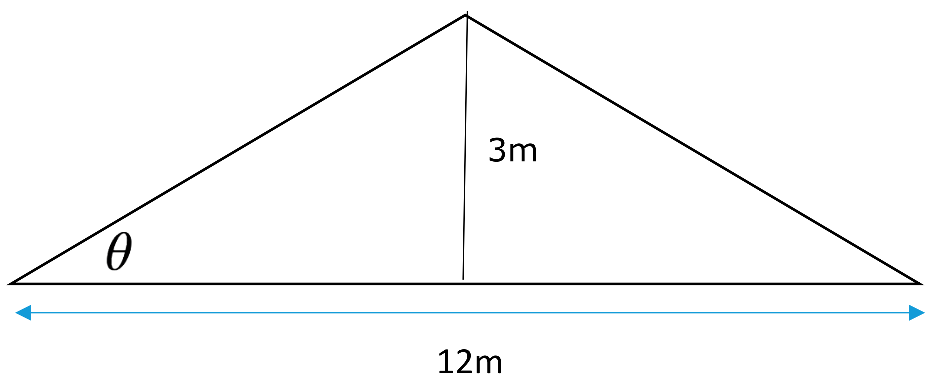
Q2.

A roof truss spans 22m and is angled at 26º. Using the measurement of the ridge height from the previous question, apply Pythagoras theorem to calculate the length of the rafter.

Q3.

The distance between the outside wall plates (span) is 12m. The rise is 3m.

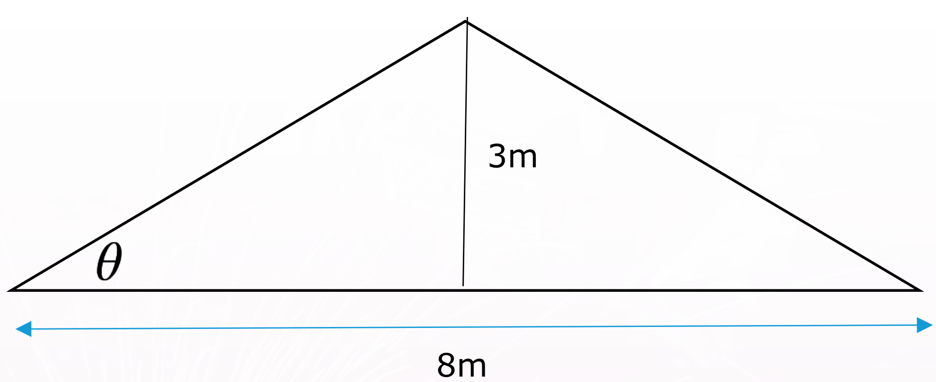
a. Find the length of the rafter on the left side. Use Pythagoras Theorem.
b. Find the pitch of the roof.



Q4.

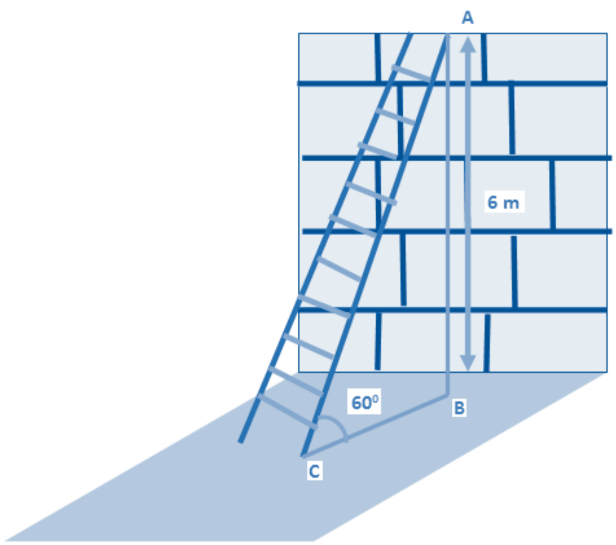
The distance between the outside wall plates (span) is 8m. The rise is 3m.

a. Find the length of the rafter on the left side. Use Pythagoras Theorem.
b. Find the pitch of the roof.



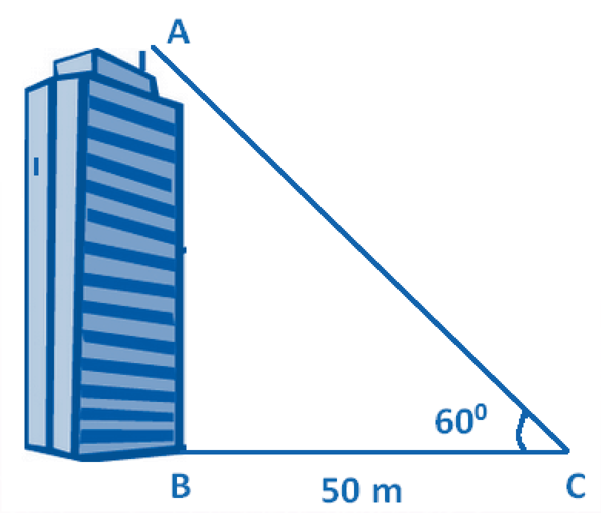
Q5.

A ladder reaches a height of 6 m from the ground (B to A) and is inclined at an angle of 60 degrees. Find the distance from the foot of the ladder to the base of the wall (C to B).



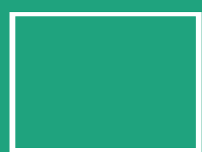
Q6.

Using Pythagoras Theorem, calculate the height (A to B) of the building below:



Formulae

Square



$A = l^2$

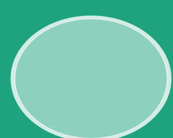
l = length of sind

Rectangle

$A = w \times h$

w = width
h = height

Circle



$A = \pi r^2$
$P = 2\pi r$

r = radius
P = perimeter

Triangle

$$A = \frac{b \times h}{2}$$

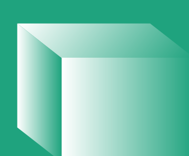
b = base
h = height

Cylinder

$V = \pi r^2 \times h$

r = radius
h = height

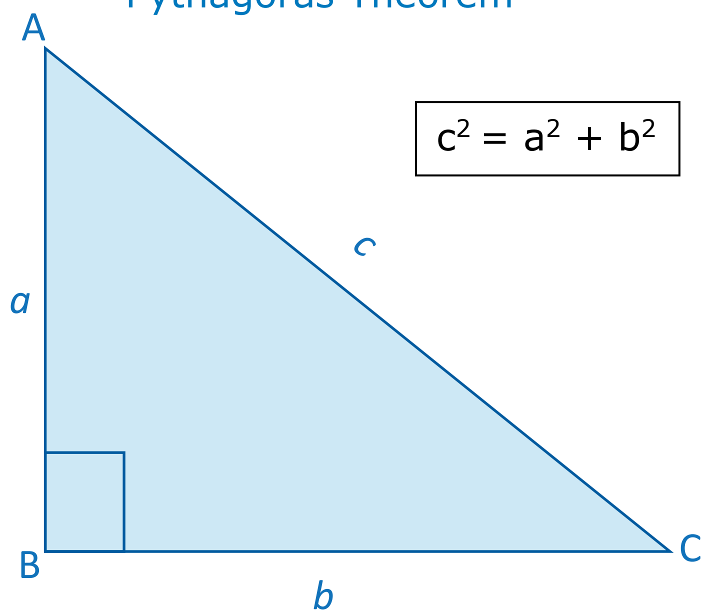
Cube



$V = s^3$

s = side

Pythagoras Theorem

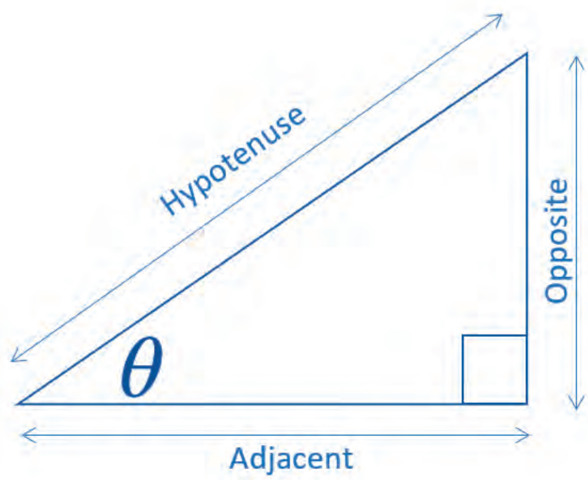


Trigonometry
SOHCAHTOA

$$\sin \emptyset = \frac{\textit{opposite}}{\textit{hypotenuse}}$$

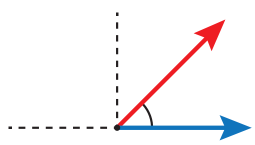
$$\cos \emptyset = \frac{\textit{adjacent}}{\textit{hyptenuse}}$$

$$\tan \emptyset = \frac{\textit{opposite}}{\textit{adjacent}}$$



Glossary of Terms

Acute: This is an angle that is greater than 0° but less than 90°.



Acute angle

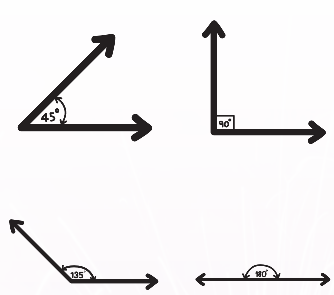
less than 90°

Algebra: A branch of mathematics that uses symbols or letters to represent variables, values or numbers, which can then be used to express operations and relationships and to solve equations.

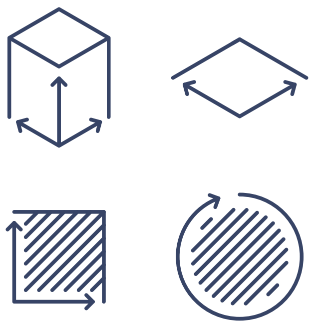
Algebraic Expression: A combination of numbers and letters equivalent to a phrase in language, e.g. $x^2 + 3x - 4$.

Algebraic equation: a combination of numbers and letters equivalent to a sentence in language, e.g. $y = x^2 + 3x - 4$.

Angle: This is made when two-line segments meet at a point (vertex), or when two lines intersect. It is be measured in degrees and can be acute, right, obtuse or reflex.



Area: The amount of space occupied. We would measure the area of a room to find out how much space would need to be covered by floor covering.

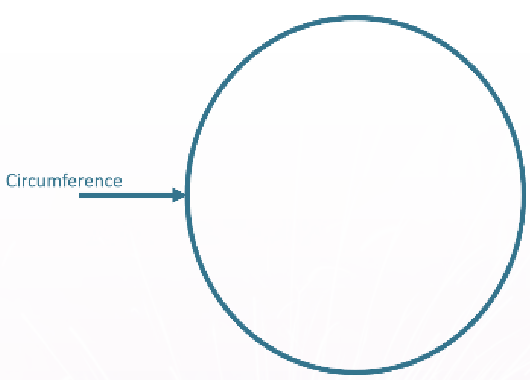


Arithmetic: The part of mathematics that studies quantity, especially as the result of combining numbers (as opposed to variables) using the traditional operations of addition, subtraction, multiplication and division.

Base *n*: The number of unique digits (including zero) that a positional numeral system uses to represent numbers, e.g. base 10 (decimal) uses 0, 1, 2, 3, 4, 5, 6, 7, 8 and 9 in each place value position; base 2 (binary) uses just 0 and 1.

Capacity: This is the internal volume of a container or simply the amount that a container can hold. Example: The capacity of the bucket is twenty litres so it takes a volume of twenty litres of water to fill it.

Circumference: This is the length of the perimeter of a circle.



Common Factor: A number that a number that can be divided into two (or more) numbers. (See factor)

Examples:

16 has factors 1,2,4,8,16

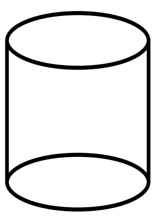
20 has factors 1,2,4,5,10,20

The common factors are: 1, 2 and 4

HCF = Highest Common Factor: This is the largest whole number that divides into two or more whole numbers. Example: The HCF of 16 and 20 is 4.

Coefficients: The factors of the terms (i.e. the numbers in front of the letters) in a mathematical expression or equation, e.g. in the expression $4x + 5y^2 + 3z$, the coefficients for x, y^2 and z are 4, 5 and 3 respectively.

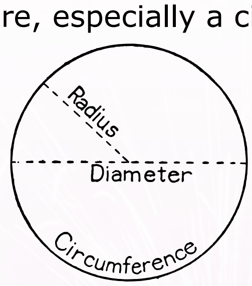
Cylinder: This is a three-dimensional shape consisting of two identical circular ends joined by one continuous curved surface.



Decimal Number: This is a real number which expresses fractions on the base 10 standard numbering system using place value, e.g. $\frac{37}{100} = 0.37$.

Denominator: This is the number below the line in a fraction.

Diameter: A straight line passing from side to side through the centre of a body or figure, especially a circle or sphere.



Exponent: The mathematical operation where a number (the base) is multiplied by itself a specified number of times (the exponent), usually written as a superscript a^n, where a is the base and n is the exponent, e.g. $4^3 = 4 \times 4 \times 4$.

Factor: A number that will divide into another number exactly, e.g. the factors of 10 are 1, 2 and 5.

Formula: A rule or equation describing the relationship of two or more variables or quantities, e.g. $A = \pi r^2$

Fraction: A way of writing rational numbers (numbers that are not whole numbers), also used to represent ratios or division, in the form of a numerator over a denominator, e.g. $\frac{1}{5}$ (a unit fraction is a fraction whose numerator is 1)

Geometry: The part of mathematics concerned with the size, shape and relative position of figures, or the study of lines, angles, shapes and their properties.

Inverse: This means 'the opposite'. The inverse of addition is subtraction. The inverse of multiplication is division.

Line: In geometry, a one-dimensional figure following a continuous straight path joining two or more points, whether infinite in both directions or just a line segment bounded by two distinct end points.

Mean: This is the simple average of a given set of data.

The mean of 8, 7,12, 0, 3 = 8 + 7 + 12 + 0 + 3 = 30

30 ÷ 5 = 6

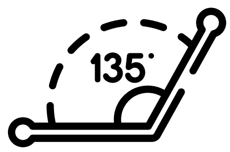
Natural Numbers: The set of positive integers (regular whole numbers), sometimes including zero.

Negative Numbers: Any integer, ration or real number which is less than 0,

e.g. -743, -1.4, -√5

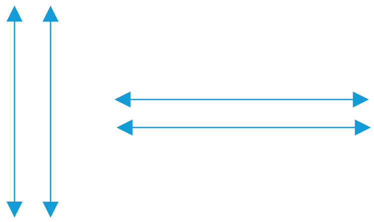
Number line: A line on which all points correspond to real numbers (a simple number line may only mark integers, but in theory all real numbers to +/- infinity can be shown on a number line).

Obtuse: This is an angle that is greater than 90° but less than 180°.

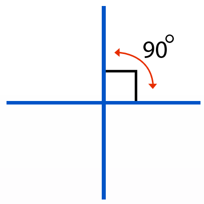


Obtuse angle of

Parallel: This is when a line runs at an equal distance apart from another line and they never meet.



Perpendicular: This is when two lines meet at right angles (90°).



Perimeter: This is the sum of the length of the sides of a figure or shape.

pi (π): The ratio of a circumference of a circle to its diameter, an irrational (and transcendental) number approximately equal to 3.141593...

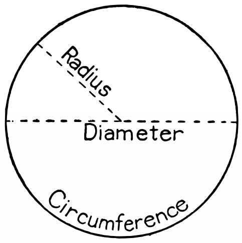
Pie Chart: This is a diagram in the shape of a circle or disc that is used to represent data. The 360° of the disc is divided in ratio into pieces of the pie.

Place Value: Positional notation for numbers, allowing the use of the same symbols for different orders of magnitude, e.g. the "one's place", "ten's place", "hundred's place", etc.

Prime numbers: integers greater than 1 which are only divisible by themselves and 1.

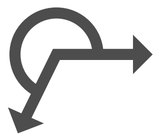
Pythagoras Theorem: The square of the hypotenuse of a right-angled triangle is equal to the sum of the squares of the two sides ($a^2 + b^2 = c^2$).

Radius: This is a line joining the centre of a circle to the edge of the circle. It is half the diameter in length.

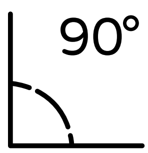


Ratio: This is a comparison of two or more quantities. Example: When making concrete you mix 9 parts of gravel with 2 parts cement. The ratio of gravel to cement is 9:2.

Reflex Angle: This is an angle that is greater than 180° but less than 360°



Right Angle: This is a 90° Angle:



Square Number: The product of a number multiplied by itself.

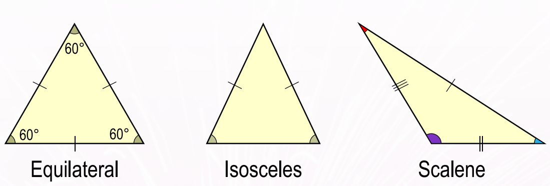
e.g: 4 x 4 = 16, so 4^2 = 16

Square Root: A number which produces a specified quantity when multiplied by itself.

e.g: 7 x 7 = 49 so ✓49 = 7

Theorem: A mathematical statement or hypothesis which has been proved on the basis of previously established theorems and previously accepted axioms, effectively the proof of the truth of a statement or expression.

Triangle: This is a three-sided shape. Example: An equilateral triangle had 3 sides of equal length, an isosceles triangle has 2 equal sides and a scalene triangle has no sides of equal length.



Glossary Adapted from:

- https://www.storyofmathematics.com/glossary.html
- https://www.ncca.ie/media/1404/glossary_of_mathematical_terms.pdf

Useful Websites:

Google Images

- Google Images is a Google's search engine for images. Instead of looking for text, you can search for pictures It's very useful for finding maths-related images

Fact Monster

- Fact Monster is an educational reference site whose broad range of content includes an encyclopedia, dictionary, thesaurus, atlas, a homework center, and educational games

Project Maths

- Includes a wide range of resources to help you with your maths and has videos and PowerPoints which help explain how to use your calculator

Free Math Help

- (American say "Math" instead of "Maths") Find helpful math lessons, games, calculators, and more. Also has a forum where you can ask maths questions

Math Is Fun

- Maths explained in easy language, plus puzzles, games, worksheets and an illustrated dictionary

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